Finite fields and combinatorial designs

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

17.1 Polynomials and polynomial functions

Fix a nonzero field F. An element of $F[x]$ is a finite formal sum $a_0 + a_1 x + \cdots + a_d x^d$. Two polynomials are equal exactly when all their coefficients agree. The symbol x is an indeterminate. Addition is coefficientwise, and multiplication uses convolution:

$$[x^k](fg) = \sum_{i+j=k} a_i b_j.$$

The degree of a nonzero polynomial is the largest index with a nonzero coefficient. Set $\deg 0 = -\infty$. Nonzero constants have degree 0. Since the product of the leading coefficients is nonzero,

$$\deg(fg) = \deg f + \deg g \qquad (f, g \neq 0).$$

Thus $F[x]$ is an integral domain, and its units are precisely the nonzero constants.

Over a finite field, distinct polynomials can define the same function. In $\mathbb{F}_2[x]$, the polynomial $x^2 + x$ is nonzero but evaluates to 0 at both 0 and 1. Equality of polynomials, congruence modulo a polynomial, and equality of evaluation functions are distinct relations.

17.2 The division algorithm

For $a, b \in F[x]$ with $b \neq 0$, there are unique q, r such that

$$a = bq + r, \qquad r = 0 \text{ or } \deg r < \deg b.$$

For existence, initialize $q = 0, r = a$. While $r \neq 0$ and $\deg r \geq \deg b$, set

$$t = \frac{\operatorname{lc}(r)}{\operatorname{lc}(b)} x^{\deg r - \deg b}, \qquad q \leftarrow q + t, \quad r \leftarrow r - tb.$$

The invariant $a = bq + r$ is preserved. Each update cancels the leading term, so the degree strictly decreases and the process terminates.

For uniqueness, suppose $bq + r = bq' + r'$. If $q \neq q'$, then $b(q - q') = r' - r$ has degree at least $\deg b$ on the left and smaller degree on the right, a contradiction. Hence $q = q'$ and $r = r'$.

The field assumption supplies $\operatorname{lc}(b)^{-1}$. In $\mathbb{Z}[x]$, dividing x by $2x$ cannot produce an integer-coefficient quotient and a remainder of degree below 1.

17.3 Division with two cancellation steps

In $\mathbb{F}_5[x]$, take $a = x^3 + 2x^2 + 4$ and $b = x^2 + 2$.

Step	Cancellation term t	Accumulated quotient q	New remainder r
Initial	None	0	$x^3 + 2x^2 + 4$
1	x	x	$2x^2 - 2x + 4 = 2x^2 + 3x + 4$
2	2	$x + 2$	$3x$

The second step subtracts $2(x^2 + 2) = 2x^2 + 4$, so the constant term cancels. The final degree is $1 < 2$, and division stops. Multiplying back gives

$$(x + 2)(x^2 + 2) + 3x = x^3 + 2x^2 + 5x + 4 \equiv x^3 + 2x^2 + 4 \pmod 5.$$

Each subtraction takes place in the coefficients modulo 5. No numerical value is substituted for x.

17.4 The factor theorem and the root bound

Evaluation $ev_\alpha(f) = f(\alpha)$ preserves addition, multiplication, and 1, so it is a ring homomorphism. Division by the monic linear polynomial $x - \alpha$ leaves a constant remainder r. Substituting α gives $r = f(\alpha)$, and therefore

$$f(\alpha) = 0 \iff (x - \alpha) \mid f.$$

A nonzero degree-d polynomial over a field has at most d distinct roots. Prove this by induction. If there is no root, the bound holds. Otherwise choose a root α and write $f = (x - \alpha)g$, where $\deg g = d - 1$. Any other root $\beta \neq \alpha$ satisfies $(\beta - \alpha)g(\beta) = 0$. Cancellation gives $g(\beta) = 0$, so there are at most $d - 1$ other roots.

The bound also holds in an integral domain. It can fail in a ring with zero divisors. In $\mathbb{Z}_8[x]$, $x^2 - 1$ has the four roots 1, 3, 5, and 7, although its degree is 2.

17.5 Polynomial gcd and Bézout coefficients

For a, b not both zero, repeatedly divide:

$$r_{i-1} = q_i r_i + r_{i+1}, \qquad \deg r_{i+1} < \deg r_i.$$

Decreasing degrees guarantee termination. A polynomial divides both r_{i-1}, r_i exactly when it divides both r_i, r_{i+1}. Thus the monic scalar multiple of the last nonzero remainder is the unique monic gcd.

Keep representations $r_i = s_i a + t_i b$ at the same time. The updates are

$$s_{i+1} = s_{i-1} - q_i s_i, \qquad t_{i+1} = t_{i-1} - q_i t_i.$$

Starting with $a = 1a + 0b$ and $b = 0a + 1b$ gives

$$\gcd(a, b) = sa + tb.$$

The final scalar used to make the gcd monic must also multiply s, t.

Consequently, the gcd is 1 if and only if some s, t satisfy $sa + tb = 1$. This algorithm will compute inverses in quotient fields.

17.6 Irreducibility and low degrees

Use the algebra convention from Chapter 14. An irreducible element is nonzero and is not a unit. In $F[x]$, it has degree at least 1, and every factorization $p = ab$ has a nonzero constant factor. Constant units are not irreducible. Every linear polynomial is irreducible.

A polynomial of degree 2 or 3 is reducible if and only if it has a root in the field. A root gives a linear factor by the factor theorem, with a remaining factor of positive degree. Conversely, a factorization into two positive degrees must include degree 1, which gives a root.

In $\mathbb{F}_2[x]$, $x^3 + x + 1$ evaluates to 1 at both 0 and 1, so it is irreducible. However,

$$x^4 + x^2 + 1 = (x^2 + x + 1)^2$$

has no root at 0 or 1 but is reducible. For degree at least 4, absence of roots excludes linear factors, but does not exclude a product of two quadratics.

17.7 Gauss's lemma

An integer polynomial is primitive when its coefficients have gcd 1. The product of two primitive polynomials is primitive. Otherwise some prime p would divide every coefficient of their product. Reduction modulo p would give a zero product in the integral domain $\mathbb{F}_p[x]$, so one factor would be zero modulo p, contrary to primitiveness.

Every nonzero rational polynomial is a rational constant times a primitive integer polynomial. Clear its denominators and divide out the gcd of the coefficients. If primitive $f \in \mathbb{Z}[x]$ factors over $\mathbb{Q}[x]$ into positive degrees, write $f = rGH$, with G, H primitive and $r \in \mathbb{Q}^\times$. Their product is primitive. Comparing the common denominator and coefficient gcd forces $r = \pm 1$. Hence the factorization can be taken over $\mathbb{Z}[x]$.

Thus a primitive integer polynomial is reducible over $\mathbb{Q}[x]$ if and only if it factors into two positive-degree polynomials over $\mathbb{Z}[x]$.

17.8 Eisenstein's criterion

Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ be primitive. Suppose a prime p satisfies

$$p \nmid a_n, \quad p \mid a_0, \ldots, a_{n-1}, \quad p^2 \nmid a_0.$$

If f were reducible over $\mathbb{Q}[x]$, Gauss's lemma would give $f = GH$ with positive-degree integer factors. Modulo p, the leading coefficients remain nonzero, so both factor degrees are preserved and their product is $a_n x^n$.

In $\mathbb{F}_p[x]$, x divides a product if and only if at least one factor has constant term zero. Repeatedly factoring out x shows that every divisor of a monomial is a monomial. Thus $\bar{G}, \bar{H}$ are positive-degree monomials. Both have constant term zero, giving $p \mid G(0), H(0)$, and hence $p^2 \mid a_0$, a contradiction.

For example, $x^4 + 2x + 2$ satisfies the criterion with $p = 2$, so it is irreducible over $\mathbb{Q}[x]$. This does not assert irreducibility over fields of every characteristic.

17.9 When a polynomial quotient is a field

Let $p \in F[x]$ be nonzero and nonconstant, of degree d. Each class modulo p has a unique representative of degree below d. Add or multiply representatives and reduce by division. The ideal property of (p) makes these operations well-defined.

If p is irreducible, a representative f of a nonzero class satisfies $0 \le \deg f < d$, so $\gcd(f, p) = 1$. Bézout gives $uf + vp = 1$. In the quotient, $[u][f] = [1]$, so every nonzero element has an inverse and the quotient is a field.

Conversely, if $p = ab$ with both factor degrees between 1 and $d - 1$, then $[a], [b] \neq 0$ but $[a][b] = 0$. The quotient cannot be a field. Therefore

$$F[x]/(p) \text{ is a field} \iff p \text{ is irreducible.}$$

If $F = \mathbb{F}_q$, the d free coefficients give q^d classes, whether or not the quotient is a field.

17.10 Coordinates and multiplication in GF(4)

Take the irreducible polynomial $x^2 + x + 1 \in \mathbb{F}_2[x]$ and let $\alpha = [x]$. Then

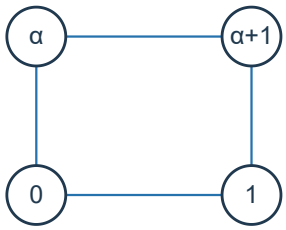
$$GF(4) = \{0, 1, \alpha, \alpha + 1\}, \qquad \alpha^2 = \alpha + 1.$$

Add coefficients modulo 2. For multiplication, expand and then replace α^2.

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17.10 Coordinates and multiplication in GF(4) (continued)

GF(4): two coefficients mod 2, with multiplication given by α² = α + 1



Horizontal: constant; vertical: α.

Addition: α + (α+1) = 1

Multiplication: α(α+1) = α²+α = 1

Multiply by α: 1 → α → α+1 → 1

Each nonzero element has additive order 2.

The two coefficients determine equality. GF(4) differs from integers mod 4: 1+1=0.

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Since $\alpha^3 = \alpha(\alpha + 1) = 1$, the nonzero elements form a multiplicative cycle of length 3. Under addition, $z + z = 0$, so each nonzero element has additive order 2. In $\mathbb{Z}_4$, the element 1 instead has additive order 4.

17.11 Complete operation tables for GF(4)

$+$	**0**	**1**	α	$\alpha+1$
0	0	1	α	$\alpha+1$
1	1	0	$\alpha+1$	α
α	α	$\alpha+1$	0	1
$\alpha+1$	$\alpha+1$	α	1	0

$\cdot$	**0**	**1**	α	$\alpha+1$
0	0	0	0	0
1	0	1	α	$\alpha+1$
α	0	α	$\alpha+1$	1
$\alpha+1$	0	$\alpha+1$	1	α

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17.11 Complete operation tables for GF(4) (continued)

For example, $(\alpha+1)^2 = \alpha^2 + 2\alpha + 1 = (\alpha+1) + 1 = \alpha$. The inverse table is

$$1^{-1} = 1, \qquad \alpha^{-1} = \alpha + 1, \qquad (\alpha+1)^{-1} = \alpha.$$

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17.12 A complete Euclidean inverse in GF(8)

Take $p = x^3 + x + 1$, already proved irreducible by its lack of roots. The quotient field has $2^3 = 8$ elements. To find $[x^2]^{-1}$, divide:

$$p = x \cdot x^2 + (x + 1), \qquad x^2 = (x + 1)(x + 1) + 1.$$

Subtraction equals addition in characteristic 2. Back-substitution gives

$$\begin{aligned} 1 &= x^2 + (x + 1)(x + 1) \\ &= x^2 + (x + 1)(p + x \cdot x^2) \\ &= (x^2 + x + 1)x^2 + (x + 1)p. \end{aligned}$$

Thus $[x^2]^{-1} = [x^2 + x + 1]$. Direct multiplication gives $x^4 + x^3 + x^2$. Using $x^3 = x + 1$ and $x^4 = x^2 + x$ reduces this product to 1.

The remainders, Bézout coefficients, and multiplication check all compute the same inverse.

17.13 Every finite field has prime-power size

In a finite nonzero field K, the sequence $0, 1, 1 + 1, \ldots$ repeats. Let p be the least positive integer with $p1 = 0$. If $p = ab$ with $1 < a, b < p$, then $a1, b1$ would be nonzero with product zero. Hence p is prime.

The map $\mathbb{Z} \to K$, $k \mapsto k1$, has kernel $p\mathbb{Z}$. Its image is $\mathbb{F}_p$, the prime subfield. View K as a vector space over $\mathbb{F}_p$. Since K is finite, it has a finite basis $e_1, \ldots, e_n$, and every element has unique coordinates

$$a_1 e_1 + \cdots + a_n e_n, \qquad a_i \in \mathbb{F}_p.$$

Therefore $|K| = p^n$ with $n \geq 1$.

This proves necessity. To prove existence for every p^n, we cannot assume without proof that an irreducible polynomial of every prescribed degree exists.

17.14 Splitting a polynomial in finitely many extensions

Every nonconstant polynomial f over a field E has an irreducible factor h. Keep factoring reducible factors. Positive integer degrees decrease, so the process stops. The quotient $E[x]/(h)$ is a field, and $[x]$ is a root of h, hence of f. The embedding of the constants E is injective.

In this extension, use the factor theorem to remove a factor $x - \alpha$. Repeat on the remaining positive-degree factor. Each step removes one linear factor. After at most $\deg f$ steps, a field L contains a full factorization

$$f = a \prod_i (x - \alpha_i).$$

If E is finite, adjoining a root via an irreducible degree-d quotient gives only $|E|^d$ elements. Every extension remains finite, so the final L is finite. This construction uses the quotient theorem without assuming irreducibles of any specified degree.

17.15 Existence for every prime power

Let $q = p^n$. Split $f = x^q - x$ over an extension L of $\mathbb{F}_p$. Formal differentiation gives $f' = qx^{q-1} - 1 = -1$, so there are no repeated roots. Indeed, $(x - a)^2 \mid f$ would force $f'(a) = 0$. Thus f has exactly q distinct roots in L.

Set $K = \{a \in L : a^q = a\}$. In characteristic p, every $\binom{p}{j}$ for $0 < j < p$ is divisible by p. Hence $(a + b)^p = a^p + b^p$, and iteration gives $(a + b)^q = a^q + b^q$.

The set K contains 0 and 1 and is closed under addition, additive inverses, and multiplication. For nonzero $a \in K$,

$$(a^{-1})^q = (a^q)^{-1} = a^{-1},$$

so it is closed under inverses as well. Therefore K is a field with exactly q elements, denoted $GF(q)$.

Together with necessity, this shows that finite field sizes are exactly the prime powers. It does not identify $\mathbb{Z}_{p^n}$ with a field.

17.16 Uniqueness up to isomorphism

Suppose $|K| = |K'| = q = p^n$. Both contain $\mathbb{F}_p$. Lagrange's theorem in the multiplicative group gives $a^{q-1} = 1$ for nonzero a. Including 0, every element is a root of $x^q - x$, so this polynomial splits in both fields.

Start with the identity embedding of $\mathbb{F}_p$. Suppose an intermediate subfield $E \subseteq K$ has been embedded in K'. Choose $a \in K \setminus E$. Finite-dimensional linear dependence among $1, a, a^2, \ldots$ gives a monic minimal polynomial $h \in E[x]$. Division shows that every polynomial vanishing at a is divisible by h, and h is irreducible.

Since $h \mid x^q - x$, transporting its coefficients through the embedding gives $\tilde{h}$ dividing the split polynomial $x^q - x$ in $K'[x]$. Choose a root $b \in K'$. The irreducible quotient gives an embedding

$$\sum c_i a^i \longmapsto \sum \varphi(c_i) b^i$$

of $E(a)$, extending the original embedding.

17.16 Uniqueness up to isomorphism (continued)

Each step strictly enlarges a finite subfield. After finitely many steps this embeds all of K into K'. Equal cardinalities make the injection bijective, hence an isomorphism. Different presentations need not give different finite fields up to isomorphism.

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17.17 An element attaining the exponent

In a finite abelian group, let e be the least common multiple of all element orders. For each prime-power factor ℓ^r of e, choose an element a whose order contains this maximal power. If $o(a) = \ell^r s$ with $\gcd(\ell, s) = 1$, then a^s has order ℓ^r.

Multiply these elements for the different primes. They commute and have pairwise coprime orders. The product has order equal to the product of those orders. For two factors, if $(ab)^t = 1_G$, then $a^t = b^{-t}$. The order of this common element divides both coprime orders, so it is 1. Thus both orders divide t. Their product is also an exponent annihilating ab, proving the claim. Induct on the number of factors.

Consequently, some element has order exactly e. The argument depends on commutativity and cannot be applied to arbitrary nonabelian groups without further hypotheses.

17.18 The multiplicative group is cyclic

For $K = GF(q)$, the group $K^\times$ is abelian of order $q - 1$. Its exponent e divides $q - 1$. All $q - 1$ nonzero elements satisfy $x^e - 1 = 0$, so the polynomial root bound gives

$$q - 1 \le e \le q - 1.$$

Hence $e = q - 1$. The preceding construction supplies an element α of order $q - 1$. Such a primitive element generates $K^\times$.

Now $\mathbb{F}_p[\alpha]$ contains 0 and every nonzero element, so it is K. The minimal polynomial h of α is irreducible, and its quotient field has size $p^{\deg h} = q = p^n$. Thus $\deg h = n$. This also proves that irreducible polynomials of every degree $n \ge 1$ exist over $\mathbb{F}_p$.

When $q = 2$, the multiplicative group has only its identity and has order 1. It is still cyclic, with primitive element 1.

17.19 An irreducible root need not be primitive

In $\mathbb{F}_3[x]$, the values of x^2+1 at 0, 1, 2 are 1, 2, 2. It is irreducible. Let $\alpha^2 = -1 = 2$ in its nine-element quotient field. Then $\alpha^4 = 1$ and $\alpha^2 \neq 1$, so α has multiplicative order 4 rather than 8.

Instead take $\beta = 1 + \alpha$. Then

$$\beta^2 = 1 + 2\alpha + \alpha^2 = 2\alpha, \qquad \beta^4 = (2\alpha)^2 = 2, \qquad \beta^8 = 1.$$

Since $\beta^4 \neq 1$, its order is 8, and it generates all nonzero elements.

Exponent	0	1	2	3	4	5	6	7
β^i	1	$1+\alpha$	2α	$1+2\alpha$	2	$2+2\alpha$	α	$2+\alpha$

For example, $\beta^{-1} = \beta^7 = 2 + \alpha$. Direct multiplication gives $(1+\alpha)(2+\alpha) = 2 + 3\alpha + \alpha^2 = 1$.

17.20 Latin squares and orthogonality

A Latin square of order n is an $n \times n$ array in which every row and every column contains each of the n symbols exactly once. A reduced square has both its first row and first column in a fixed order. Having only the first row in that order is first-row normalization, a weaker condition.

The array $L(x, y) = x + y \pmod{n}$ is Latin for every $n \geq 1$. Fixing one coordinate makes translation in the other coordinate a permutation. Other Latin squares need not be cyclic shifts or satisfy associativity.

Two Latin squares L, M are orthogonal if the n^2 ordered pairs $(L(x, y), M(x, y))$ list all symbol pairs exactly once. Common row and column permutations preserve orthogonality. Independently renaming the symbols in each square also preserves it. Permuting the rows of only one square can destroy it.

17.21 Finite fields construct a complete set of MOLS

Let $F = GF(q)$. For each nonzero a, define

$$L_a(x, y) = ax + y, \qquad x, y \in F.$$

For fixed x, this is translation in y. For fixed y, multiplication by the nonzero a is invertible. Thus each L_a is Latin.

For $a \neq b$, prescribe a symbol pair (u, v) and solve

$$ax + y = u, \qquad bx + y = v.$$

The unique solution is

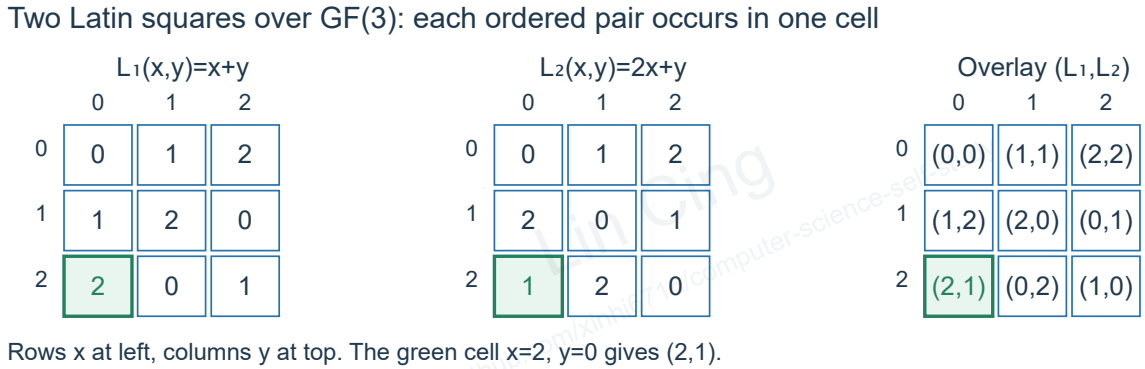
$$x = (a - b)^{-1}(u - v), \qquad y = u - ax.$$

Every pair occurs exactly once, so L_a, L_b are orthogonal. All $a \in F^\times$ give $q - 1$ mutually orthogonal Latin squares, or MOLS.

The proof needs both a and $a - b$ to be invertible. In $\mathbb{Z}_4$, copying L_1, L_3 fails because their coefficient difference 2 is not invertible. The fact that each square is Latin does not ensure their orthogonality.

17.22 Locating an ordered pair

In $GF(3)$, the nonzero coefficients are 1 and 2. In all three panels below, rows are x and columns are y. The third panel superimposes the two squares cell by cell without moving positions.



To locate the ordered pair $(2, 1)$, compute

$$x = (1 - 2)^{-1}(2 - 1) = 2^{-1} \cdot 1 = 2, \qquad y = 2 - 1 \cdot 2 = 0 \pmod 3.$$

17.22 Locating an ordered pair (continued)

It lies in the green cell $x = 2, y = 0$. All nine ordered pairs are distinct, as predicted by the general unique-solution proof.

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17.23 The upper bound on the number of MOLS

Let $n \geq 2$, and suppose there are k pairwise orthogonal Latin squares. Rename the symbols separately in each square so that its first row is $0, 1, \ldots, n-1$. Symbol renaming preserves cell correspondence and orthogonality.

Examine the entry in the second row and first column of each square. It cannot be 0, because that column already has 0 in its first row. If two squares had the same entry $a \neq 0$ in this cell, their ordered pair (a, a) would occur twice. The other occurrence is in column a of the first row, contradicting orthogonality.

These k entries must therefore be distinct members of the $n-1$ nonzero symbols. Hence $k \leq n-1$. The finite-field construction attains the bound.

Only first-row normalization was used. Independently permuting rows to make every square reduced could change their alignment and invalidate the proof.

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17.24 Affine planes and field coordinates

An affine plane has points, lines, and incidence satisfying three axioms. Two distinct points determine exactly one line. For a line ℓ and a point P off it, exactly one line through P is disjoint from ℓ. At least three points are noncollinear.

Over $F = GF(q)$, take points F^2 and lines $x = c$ or $y = mx + b$. Two points with equal x-coordinates determine their unique vertical line. Otherwise their unique slope is

$$m = (y_2 - y_1)(x_2 - x_1)^{-1},$$

and $b = y_1 - mx_1$ gives the unique intercept.

Distinct lines of the same slope are disjoint. Different slopes give a unique intersection by solving the two equations, and a vertical line meets a nonvertical line exactly once. Thus the unique parallel through an external point has the same direction, with intercept fixed by the point. The points $(0, 0), (1, 0), (0, 1)$ are noncollinear, proving nondegeneracy.

17.25 Directions, parallel classes, and parameters

Each nonvertical direction $m \in F$ has q lines $y = mx + b$. Including the vertical direction gives $q + 1$ parallel classes. The q lines of one class are pairwise disjoint and partition all q^2 points. Lines from different classes meet in exactly one point.

The affine plane $AG(2, q)$ therefore has parameters

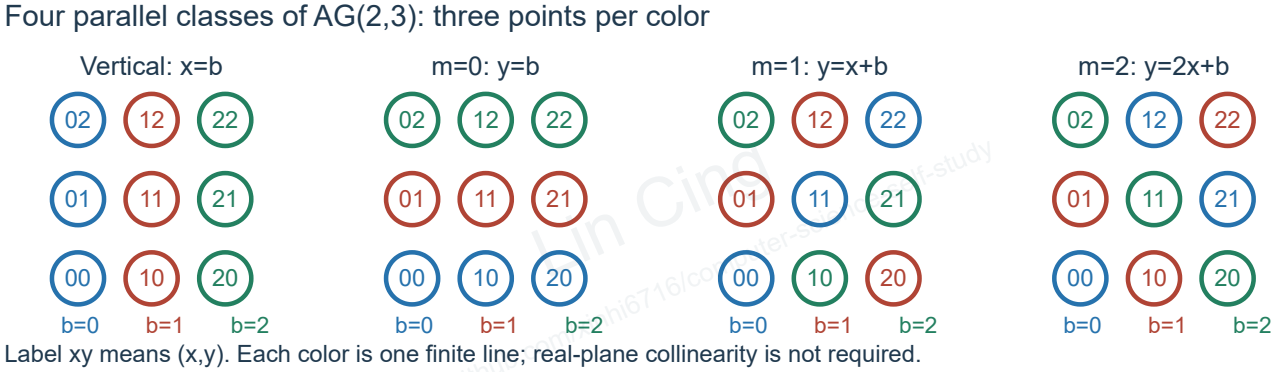
$$v = q^2, \qquad b = q(q + 1), \qquad k = q, \qquad r = q + 1.$$

A vertical line allows any y; every other line allows any x. Thus each line has q points. Exactly one line from each direction passes through a given point, so $r = q + 1$.

Parallel classes partition the set of all lines. Within a single class, the lines partition the set of all points. These are two different partitions.

17.26 All twelve lines of AG(2,3)

Each panel has the same nine points over $GF(3)$; colors distinguish $b = 0, 1, 2$. Label $12 = (1, 2)$, not decimal twelve, satisfies $2 = 2 \cdot 1 + 0$. Incidence follows field equations, not real-plane collinearity.



For $m = 2$, the blocks are $b = 0 : \{00, 12, 21\}$, $b = 1 : \{01, 10, 22\}$, and $b = 2 : \{02, 11, 20\}$.

17.27 Complete MOLS and affine planes

Fix $n \geq 2$. From $n - 1$ MOLS of order n, take the n^2 cells as points. Rows and columns give two parallel classes. In each square, cells with the same symbol form lines in one more class. There are $n + 1$ classes. An order-1 Latin square can be defined, but its single cell cannot give a nondegenerate affine plane.

Each class partitions the cells. Lines of different classes meet exactly once. A row meets a column in its unique cell. A row or column meets a symbol line once by the Latin property. Symbol lines from different squares meet once by orthogonality.

Fix a cell P. Its row and column cover $2(n - 1)$ other cells. Each square's same-symbol line covers $n - 1$ more. Different squares cannot overlap there, since that would repeat an ordered pair. The total is

$$2(n - 1) + (n - 1)^2 = n^2 - 1.$$

Thus every other point shares exactly one line with P. The same class supplies the unique parallel through an external point, giving an affine plane.

17.27 Complete MOLS and affine planes (continued)

Conversely, choose two directions in an affine plane of order n as rows and columns. Their intersections give cells. In each of the other $n-1$ directions, use line labels as symbols. Unique intersections give the Latin property and orthogonality. This correspondence does not claim that every abstract affine plane has field coordinates.

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17.28 Designs and general counting formulas

A $t-(v,k,\lambda)$ design has v points and indexed blocks, each a k-subset. Every t-subset lies in exactly λ blocks. Assume $1 \le t \le k \le v$ and $\lambda \ge 1$. Unless the design is called simple, different block indices may carry the same subset, and counts include multiplicity.

Count pairs (T, B) with $T \subseteq B$ and $|T| = t$:

$$b\binom{k}{t} = \lambda\binom{v}{t}, \qquad b = \lambda\binom{v}{t}/\binom{k}{t}.$$

For any fixed s-subset S, with $0 \le s \le t$, count $S \subseteq T \subseteq B$:

$$\lambda_s\binom{k-s}{t-s} = \lambda\binom{v-s}{t-s}.$$

The number of blocks containing S therefore depends only on s, not on the choice of S. In particular, $\lambda_0 = b$ and $\lambda_1 = r$. All these values must be integers. These are necessary conditions for existence.

17.29 Two counts for a BIBD

A nondegenerate balanced incomplete block design, or BIBD, is a $2-(v,k,\lambda)$ design with $2 \le k < v$ and $\lambda \ge 1$. Each point lies in exactly r blocks.

Counting point-block incidences gives

$$vr = bk.$$

Fix a point P and count (Q, B) with $Q \neq P$ and $P, Q \in B$. Choosing the block first gives $r(k-1)$, while choosing Q first gives $\lambda(v-1)$. Hence

$$r(k-1) = \lambda(v-1).$$

For $2-(7,3,1)$, this gives $r = 6/2 = 3$ and $b = 7 \cdot 3/3 = 7$. The affine plane $AG(2,q)$ is a $2-(q^2,q,1)$ design. Substitution gives $r = q+1, b = q(q+1)$, agreeing with the direction count.

Parameter conventions vary. The five-parameter notation v, b, r, k, λ must not be read as the notation $t-(v,k,\lambda)$.

17.30 Fisher's inequality

Let M be the $v \times b$ incidence matrix, with points as rows and indexed blocks as columns. Each row has r ones, and two different rows have λ common ones. Over the real numbers,

$$MM^T = (r-\lambda)I + \lambda J.$$

Since $r = \lambda(v-1)/(k-1) > \lambda$, every nonzero real vector z satisfies

$$z^T MM^T z = (r-\lambda)\sum_i z_i^2 + \lambda\left(\sum_i z_i\right)^2 > 0.$$

Thus MM^T has rank v, and

$$v \le \operatorname{rank} M \le b.$$

This proves Fisher's inequality $b \ge v$.

The proposed design $2-(16,6,1)$ gives integral values $r = 3, b = 8$, but $8 < 16$. It therefore does not exist. This explicit obstruction shows why divisibility alone is insufficient.

17.31 Projective plane axioms

A projective plane satisfies three axioms. Any two distinct points lie on exactly one line. Any two distinct lines meet in exactly one point. There are four points with no three collinear. The last axiom excludes degenerate incidence structures. A finite plane has order $q \geq 2$ when every line has $q + 1$ points.

In the field model $PG(2, q)$, points are one-dimensional subspaces $[v] = \{cv : c \in F_q\}$ of F_q^3, with $v \neq 0$. Lines are two-dimensional subspaces, and incidence is containment. The vectors v, cv, with $c \neq 0$, represent the same point and must not be counted separately.

Distinct points have independent representative vectors. Their span is the unique two-dimensional subspace containing them. Two distinct two-dimensional subspaces intersect in dimension 1. The dimension formula gives at least 1, while dimension 2 would make them equal. Thus two lines meet in exactly one point.

The four points $[1, 0, 0], [0, 1, 0], [0, 0, 1], [1, 1, 1]$ have every three independent, with determinant ± 1, proving nondegeneracy.

17.32 Deriving the parameters of PG(2,q)

There are $q^3 - 1$ nonzero vectors in three dimensions. Each one-dimensional subspace contains $q - 1$ nonzero vectors, and these groups are disjoint. Thus

$$v = (q^3 - 1)/(q - 1) = q^2 + q + 1.$$

A two-dimensional line contains $(q^2 - 1)/(q - 1) = q + 1$ points.

Every two-dimensional subspace is the kernel of a nonzero linear functional $a \cdot x$. Two functionals have the same kernel exactly when one is a nonzero scalar multiple of the other. Hence $b = q^2 + q + 1$. Lines through a fixed point P correspond to one-dimensional subspaces of F_q^3/P, so there are $q + 1$ of them.

The result is a $2 - (q^2 + q + 1, q + 1, 1)$ design with $v = b$ and $r = k = q + 1$.

17.32 Deriving the parameters of PG(2,q) (continued)

Duality exchanges points with lines and reverses incidence. The two uniqueness axioms exchange places. Nondegeneracy also needs proof. From four points A, B, C, D with no three collinear, take the lines AB, BC, CD, DA. No three are concurrent. For example, AB, BC meet at B, which lies on neither CD nor DA; the other triples follow cyclically. These four lines are the required four points in the dual. Thus the dual is a projective plane.

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17.33 Adding and deleting a line at infinity

Write an affine point (x, y) as $[x : y : 1]$. Add $P_m = [1 : m : 0]$ for each slope m, and $P_\infty = [0 : 1 : 0]$ for the vertical direction. All new points lie on $\ell_\infty : z = 0$.

Extend $y = mx + b$ to $y = mx + bz$, containing P_m. Extend $x = c$ to $x = cz$, containing P_∞.

Parallel affine lines meet at their unique direction point. Lines of different directions retain their affine intersection. An old point and a direction point determine the unique line of that direction through the point. Two distinct direction points determine ℓ_∞. These cases verify both projective uniqueness axioms. Adding $q + 1$ points and one line, with one new point per old line, gives the projective parameters.

Conversely, delete a projective line ℓ_∞ and its points. Two remaining lines become parallel exactly when their former intersection was on ℓ_∞. For an external point P and remaining line ℓ, let $Q = \ell \cap \ell_\infty$. The unique projective line PQ, with Q removed, is the unique parallel to ℓ through P. The unique join of two remaining points also survives.

17.33 Adding and deleting a line at infinity (continued)

For nondegeneracy, choose $P \notin \ell_\infty$, which the projective axiom guarantees. A line through P meets ℓ_∞ at T. Since it has $q+1 \geq 3$ points, choose $Q \neq P, T$ on it. Choose $S \neq T$ on ℓ_∞, and $R \neq P, S$ on PS. Both Q, R are off ℓ_∞. The lines PS, PQ meet only at P, so P, Q, R are noncollinear. All three affine axioms hold.

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17.34 Fano points, coordinates, and blocks

In $PG(2,2)$, there is no nontrivial nonzero scalar to identify. Label nonzero binary vectors by $1 = 001, \ldots, 7 = 111$. Two distinct vectors u, v span a two-dimensional subspace whose three nonzero vectors are $u, v, u+v$.

The seven line blocks are

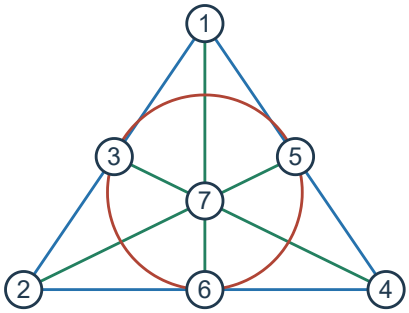
$$123, \quad 145, \quad 167, \quad 246, \quad 257, \quad 347, \quad 356,$$

where, for example, 123 means $\{1,2,3\}$. Every pair u, v has the unique third point $u+v$, so it appears in exactly one block.

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17.34 Fano points, coordinates, and blocks (continued)

Fano plane: seven points and seven blocks; the circle is a line



L1 = {1,2,3}	1 ↔ 001
L2 = {1,4,5}	2 ↔ 010
L3 = {1,6,7}	3 ↔ 011
L4 = {2,4,6}	4 ↔ 100
L5 = {2,5,7}	5 ↔ 101
L6 = {3,4,7}	6 ↔ 110
L7 = {3,5,6}	7 ↔ 111

Right: nonzero binary vectors. The three vectors in each block XOR to 000.

A line is an incidence block. The red circle 356 is also a projective line. Projective lines need not be drawn as straight lines in the real plane.

17.35 Deleting a Fano line

Delete line 123 and its points 1, 2, 3. The affine points are $\{4, 5, 6, 7\}$. Remove the point at infinity from each of the other six lines.

Point at infinity	Projective lines	Affine lines in one parallel class
1	145, 167	$\{4, 5\}, \{6, 7\}$
2	246, 257	$\{4, 6\}, \{5, 7\}$
3	347, 356	$\{4, 7\}, \{5, 6\}$

The two blocks in each class partition the four points, and every pair lies in exactly one block. This is the affine plane of order 2. Lines that previously met at a deleted point are now parallel. Each point at infinity corresponds explicitly to one parallel class.

17.36 Fano lines and Hamming codewords

A Fano line $\{u, v, w\}$ satisfies $u + v + w = 000$. If the parity-check matrix has these seven nonzero vectors as columns, putting 1 in the three corresponding positions and 0 elsewhere gives a weight-3 codeword.

The columns of H in Chapter 16 are not in binary-label order. Their order is

$$110, 101, 011, 111, 100, 010, 001,$$

corresponding to Fano labels $6, 5, 3, 7, 4, 2, 1$. For example, the ones of codeword 1110000 are in positions 1, 2, 3. They give the line $\{6, 5, 3\} = 356$, the red circle in the figure.

The seven lines correspond to the seven weight-3 codewords through the XOR of the same columns. A column permutation must be accompanied by the same permutation of code coordinates and point labels.

17.37 Conditions to check when reconstructing an argument

Reconstruct the calculations from their structures. This includes polynomial degrees and gcds, inverses in irreducible quotients, finite-field existence and isomorphism, unique solutions for MOLS, every point-line incidence case, design double counts, and the rank bound.

Finite-field constructions give planes of prime-power orders. They do not prove that every abstract plane admits field coordinates. Existence of a Latin square does not imply existence of MOLS, and integral design parameters do not supply an incidence structure.

The exercises vary the polynomials, moduli, coefficients, and blocks. They require intermediate computations and general derivations. Each has a complete English solution.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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